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Results of Long-Term Analyses of the Salt (Macro) Batch 14 Tank 21H Qualification Sample

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January 2025
SRNL-STI-2025-00012

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Printed in the United States of America

Prepared for
U.S. Department of Energy

SRNL-STI-2025-00012
Revision #0

Keywords: *SWPF, cesium*

Retention: *permanent*

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Savannah River National Laboratory is operated by Battelle Savannah River Alliance for the U.S. Department of Energy under Contract No. 89303321CEM000080.

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EXECUTIVE SUMMARY

Savannah River National Laboratory (SRNL) analyzed samples from Tank 21H in support of qualification of Salt Waste Processing Facility (SWPF) Salt Batch 14. This document reports the long-term results of the analyses of the sample of Tank 21H. Analysis of this sample indicates that the material does not display any unusual characteristics or observations, such as visually apparent solids. This report satisfies Deliverable 3 of the Technical Task Request (TTR).

TABLE OF CONTENTS

LIST OF TABLES vii
LIST OF ABBREVIATIONS viii
1.0 Introduction 1
2.0 Experimental Procedure 1
2.1 Quality Assurance 1
3.0 Results and Discussion 1
4.0 Conclusions 3
5.0 References 4

LIST OF TABLES

Table 1. Analytical Results 2

Table 2. ^{129}I Results for Past Salt Batches 3

LIST OF ABBREVIATIONS

%RSD	Percent Relative Standard Deviation
SRNL	Savannah River National Laboratory
SRMC	Savannah River Mission Completion
TTR	Technical Task Request
TTQAP	Task Technical and Quality Assurance Plan
WAC	Waste Acceptance Criteria

1.0 Introduction

This report provides the analytical laboratory results of the long term SWPF Salt Batch 14 samples from Tank 21H. These results will be used by Savannah River Mission Completion (SRMC) to show compliance with downstream facilities. This work was specified by a TTR[i] and Task Technical and Quality Assurance Plan (TTQAP).[ii] Details for the work are contained in controlled laboratory notebooks.[iii]

2.0 Experimental Procedure

A single 200 mL Tank 21H sample (HTF-21-24-111) was pulled and delivered to SRNL on November 5, 2024. The sample was a single depth sample obtained approximately 62" from the bottom of the tank (transfer pump suction). Tank 21H was mixed at full speed for approximately 72 hours by November 4, 2024 before the sample was pulled; the sample was pulled approximately 72 hours after pump shutdown. The sample was pale yellow with no apparent solids.

Samples were filtered via a 0.45 μm nylon syringe filter, apart from the samples for the methyl mercury. The methyl mercury sub-samples which were diluted with water and HCl in order to preserve the ratios of the various mercury species. All dilutions are accounted for in the reported values provided.

2.1 Quality Assurance

Requirements for performing reviews of technical reports and the extent of review are established in Manual E7, Procedure 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2. The customer requested that a Functional Classification of Safety Class apply to this work. A Design Verification technical review was performed via a document review according to the applicable elements detailed in Section 5.3.1 'Design Verification by Document Review' of E7 2.60. Data collection and analysis methods used in this work comply with this requirement as detailed in the TTQAP. Results from this report are not RW-0333P as per the TTR.

3.0 Results and Discussion

Results for the long-term analyses are listed in Table 1. Values in parentheses are the percent relative standard deviation (%RSD) values based on replicate sample analysis.

The ^{129}I analysis used a more recent analytical method that involved a chemical separation in the cells, and was run in triplicate.

Table 1. Analytical Results

Analyte	Result (pCi/mL)	WAC Requirement (pCi/mL)
^{3}H	3.78E+03 (0.0%)	5.63E+05
^{14}C	2.65E+02 (1.8%)	1.13E+05
^{60}Co	<2.64E+01	9.75E+02
^{99}Tc	1.15E+05 (13%)	2.11E+05
^{125}Sb	<3.56E+02	7.99E+03
^{126}Sn	1.57E+03 (5.5%)	1.80E+04
^{129}I	1.03E+02 (5.1%)	1.00E+02
^{144}Ce	<6.48E+02	3.12E+04
^{154}Eu	<8.10E+01	1.62E+03
^{155}Eu	<3.36E+02	none
^{232}Th	<2.20E-04	2.88E+03
^{237}Np	4.65E+00 (5.1%)	1.00E+04
^{241}Pu	5.38E+03 (4.1%)	8.38E+05
^{241}Am	<5.76E+00	6.67E+04
^{244}Cm	<1.05E+01	6.67E+04
^{245}Cm	<7.65E+00	2.25E+05
Me-Hg	8.50 (0.58%)	3.50E+02 mg/L

The analytical uncertainties for the radiochemical measurements varied from 5-20%. The analytical uncertainty for the methyl mercury analyses is 20%. The methyl mercury value is typical of previous SWPF salt batches.

SRNL will note that the ^{99}Tc and ^{129}I values are approaching or exceeding their WAC requirements. In comparison, previous batches 11 and 12 exceeded the ^{129}I limits. For context, the last five ^{129}I results from Salt Batches 9-13 are given. See Table 2.

Table 2. ^{129}I Results for Past Salt Batches

Salt Batch	SourceTank	^{129}I (pCi/mL)
9	21H	9.38E+01 (7.8%)
10	41H	9.45E+01 (3.4%)
11	21H	1.55E+02 (4.6%)
12	42H	1.40E+02 (1.4%)
13	42H	8.48E+01 (3.1%)

There is a clear pattern of being close to, or slightly exceeding the WAC requirements in recent times.

4.0 Conclusions

Analysis of the Tank 21H SWPF Salt Batch 14 sample indicates that the material does not display any unusual characteristics or observations, such as visually apparent solids. The analytical results received thus far are mostly typical of previous salt batches, and mostly consist of less-than values.

This memo satisfies Deliverable 3 of the Technical Task Request (TTR).

5.0 References

[i] J. B. Goldblatt, "Salt Batch Qualification for Feed to Salt Waste Processing Facility", X-TTR-H-00090, Rev. 4, August 2023.

[ii] T. B. Peters and J. R. Dekarske, "Task Technical and Quality Assurance Plan for SWPF Feed Batch Qualification Testing", SRNL-RP-2019-00582, Rev. 4, November 2023.

[iii] T. B. Peters, "Salt Batch 14 Qualification", ELN, A4571-00527-19.

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